

Enable actual quality measurements

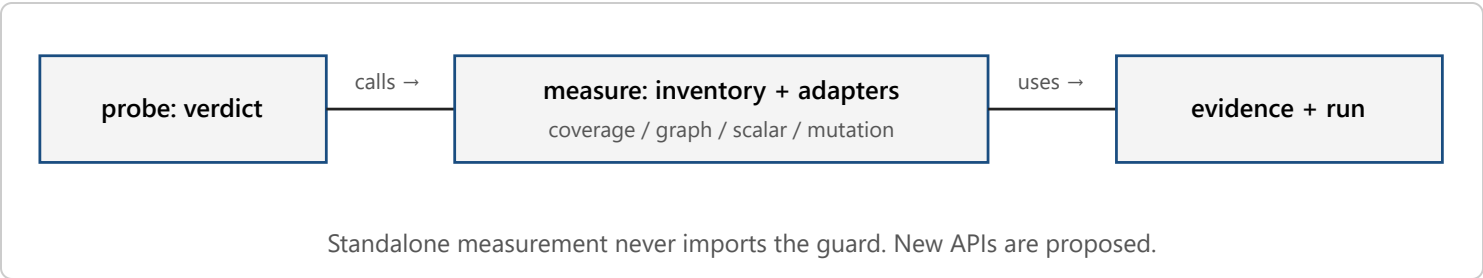
17 September 2026 · Prerequisite Q · Revision 2 · Independently accepted design
[Precise contracts, adapters and check sequence](#) · [Grounding](#) · [Approved functional continuation](#)

1. Decision and boundary

Build producers, not another gate. Current collectors cannot supply complete mixed-language proof. Q1–Q5 enable all nine metrics; actual proof still gates release. Mutation stays 60%; evaluation stays 0.9.

New autonomous policy: 100% changed executable lines and decision outcomes under the declared models. This is not historical policy or human approval. An empty executable denominator is unavailable, not 100%. Tool qualification remains required.

2. Program design



Component / criterion	Interface, change and responsibility
<code>measure.py</code> · Q1	<code>inventory(context, revision, changed_production)</code> , <code>collect_pair(context, config, base, head)</code> : lower-layer discovery owner. Probe imports it, never the reverse. Retain errors, exclusions and zero-output files.
<code>measure_graph.py</code> + <code>measure_js.mjs</code> · Q1/Q2	<code>parse_files(context, inventory, config) → ParsedInventory</code> , <code>evaluate_rules(graph, rules)</code> : Python AST/TypeScript produce shared source-bound records for graph, scalar, coverage and candidate adapters. No reverse imports/framework.
<code>measure_scalars.py</code> · Q2	<code>collect(context, parsed, config)</code> : jscpd duplication; lizard complexity; Vulture/JS dead-code findings; Ruff/TypeScript static diagnostics. Identical semantics at base/head.
<code>measure_coverage.py</code> · Q3	<code>collect(context, parsed, config, suites)</code> , <code>select(candidate, coverage, suites)</code> : Python child coverage and Node V8 ranges; union line sets, retain uncovered files.
<code>mutate.py</code> + <code>unittest_result.py</code> · Q3–Q5	Extend <code>build_mutants</code> , <code>select_test_run</code> , <code>validate_mutation_results</code> : AST/token candidates, phase-aware proof and selected suites. Stable controller executes separate target bytes, including mutated controller sources.
<code>probe.py</code> + producers · Q1–Q4	<code>validate_measurement_report(report, context, base, head, config, policy, artifacts)</code> : producers rederive all nine metrics, comparisons and verdict from validated raw evidence. Reject inconsistent summaries; no guard dependency.

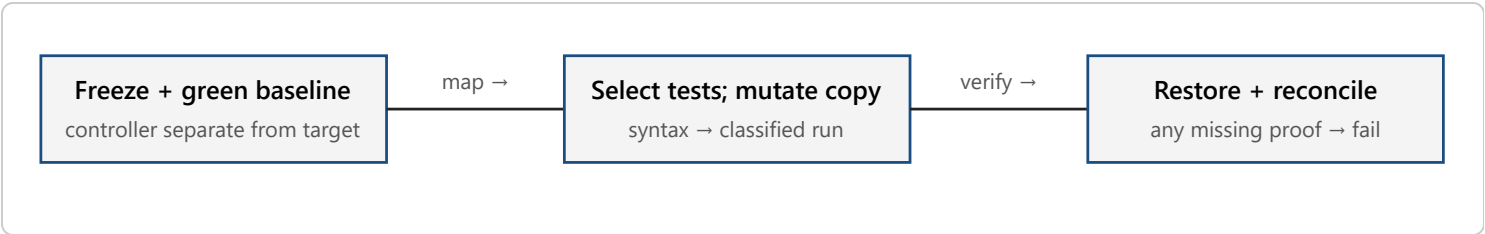
3. Exact meaning of “measured”

Inventory includes tests/examples/benchmarks; coverage/mutation select changed production. Zero findings require successful processing. Unsupported inputs remain explicit. `DirectoryRef = "."` | `Path` permits root only in directory fields; file/artifact paths remain non-root, non-escaping.

Compare cycle/architecture identities in the literal-import graph; enforce layering and forbidden test/benchmark imports. Unresolved locals or parse failures are unavailable. Computed imports/reflection/commands remain outside this static model, not proved safe.

Producer validators bind raw records, exact receipt partitions, revision/source/tool/policy and owned artifacts; recompute values, denominators and verdict. Hash identity alone is not aggregate consistency, authenticated identity or containment.

4. Self-measurement without self-corruption



Retain the deterministic 20-candidate sample and expose omissions. Coverage selects *tests*, *not mutants*; uncertain mapping and survivors use full-suite confirmation. Python kills require the actual failed test assert instruction or qualified unittest failure path, not an ancestor assertion line. Setup/application errors and timeouts never kill.

The historical Python suite took about 12 minutes. Measure durations and cost the schedule before launch; keep the controller outside mutated target files. Budget exhaustion is incomplete, never tail dropping or timeout-as-survival.

5. Incremental delivery and proof

Slice	Observable result / real tests
Q1 · shortest first	Inventory + Python AST graph + architecture/cycle reconciliation. A Python-only Git fixture measures real graph values while other required metrics still fail closed. Tests: root/nested cwd, standalone imports, failed enumeration, swapped revision, zero imports, replaced cycle, forbidden edge and forged graph/summary rejection. No external installation needed for this first slice.
Q2 · after Q1	Qualify external tools; extend every scalar to Python/JS/MJS/TS. Tests: zero-function/zero-clone files, ignored inputs, parse failure, dynamic-site disclosure, same-rule base/head and new-violation identity.
Q3 → Q4 → Q5	Phase-aware unittest, subprocess coverage and selection → mixed Python/parsed-JS mutation → actual frozen whole-change probe. Test uncovered files, both branch arms, setup/cleanup/subtests, templates/regexes, controller self-mutation and stale/forged summaries. Each slice: unit + real CLI integration, fresh verification/review; final native/corpus regression.

Q2 prerequisites: qualify skill-owned coverage.py, lizard, Vulture, Ruff, jscpd and TypeScript 5.9.3; installed tsc 7.0.2 is not that parser API. No WSL or project framework dependencies. Documented platform support is not execution proof.

Rejected: imported scores, extension-only inventories, unqualified Windows tools and controller exclusions. Q is not passed; publication's 403 remains separate. Preserved [review](#), [dispositions](#) and [independent acceptance](#).